Zili Tan

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Research Interests

Self-Improving AI Systems, Learning Dynamics, Continual Adaptation, and Multi-Agent Systems.

Academic Background

University of Amsterdam & Vrije Universiteit Amsterdam *09/2025 – 06/2027 (expected)*
MSc in Physics and Astronomy (joint degree), Theoretical Physics track Current average grade: 8.62/10

Sun Yat-sen University *09/2021 – 07/2025*
Bachelor of Science in Physics Average grade: 8.57/10

Publications & Preprints

Meng Bi, Jinhai Yang, **Zili Tan**, Bowei He, Hong Huang, and Xue Liu. *Wright–Gödel Machine: Mitigating Homogenization in Self-Improving Agents.* **Under review at ICLR 2027.**

Research Experience

Continual Learning in Resource-Constrained Agents *10/2026 – Present*
Exploratory research

- Investigating whether prior experience remains useful across changing task streams under memory and compute constraints.
- Built and locally tested a research prototype; prepared a manuscript draft.
- Designed controlled evaluations with matched resource budgets to assess cumulative task performance and the value of prior experience for recurring and held-out tasks.

Agentic AI Workflows for SMEFT Event Generation at the LHC *10/2026 – Present*
MSc Thesis, Advisor: Prof. Wouter Verkerke, Nikhef / University of Amsterdam

- Developed a single-agent workflow in LangGraph that configures MadGraph from physics requests, generates parton-level SMEFT events, and produces kinematic plots.
- Integrated mandatory configuration and sample checks with corrective feedback and review of unresolved choices in the physics setup.
- Evaluated four leading-order tasks in paired tests against independent reference samples: 4/4 produced matching samples with checks and review enabled, versus 0/4 with them disabled.
- Tested Python 3 compatibility across all ten official SMEFTsim UFO variants using 100-event MadGraph runs.
- Published a bilingual tutorial on SMEFTsim event generation and detector simulation.

Genetic Homogenization in Self-Improving Coding Agents *08/2026 – 09/2026*
Collaborative project with Meng Bi et al.

- Investigated diversity collapse in recursively self-improving coding agents across Darwin, Huxley, and Mendel Gödel Machine variants.
- Jointly introduced the concept of genetic homogenization to characterize the loss of heritable design diversity during iterative self-improvement.
- Studied the exploration–exploitation tradeoff underlying this collapse and proposed EXP3 as an alternative to Thompson Sampling for maintaining architectural diversity.
- Jointly introduced an evaluation strategy inspired by BAI-MCTS.

Observer-Relative Inflationary Histories in de Sitter JT Gravity *06/2026 – 09/2026*
Advisors: Prof. Ben Freivogel and Prof. Jan Pieter van der Schaar, University of Amsterdam

- Studied de Sitter JT (Jackiw–Teitelboim) gravity coupled to a homogeneous inflaton and a weakly coupled observer.
- Found that the current observer-relative gravitational path integral (GPI) prescription fails to normalize the no-boundary state, suggesting a possible inconsistency in the Hartle–Hawking proposal within the observer-relative Hilbert space.

Lorentzian Gravitational Path Integral in the de Sitter Static Patch *04/2026 – 06/2026*
Advisor: Prof. Jan Pieter van der Schaar, University of Amsterdam

- Reviewed how Lorentzian path-integral approaches can avoid the conformal factor problem.
- Used Claude Code to expand the Einstein–Hilbert action around the D-dimensional de Sitter static patch.
- Obtained the one-loop spectrum of de Sitter space from the Lorentzian gravitational path integral.

Holographic Cosmology and Inflationary Cosmology *10/2022 – 06/2025*
BSc Thesis, Advisor: Prof. Chengcheng Han, Sun Yat-sen University

- Studied the standard cosmological model and inflationary cosmology in Prof. Han's research group.
- Reviewed the holographic principle, holographic entropy bounds, and models proposed by Banks and Fischler.
- Analyzed how holographic models address the problem of low-entropy initial conditions and assessed prospects for testing holographic cosmology through future CMB and primordial gravitational-wave observations.

Competitions & Awards

National Training Program of Innovation for Undergraduates *12/2022 – 03/2024*
Presenter and Team Member

- Reviewed slow-roll models and compared their predicted power spectra with Planck TT observations.
- Final report rated **Excellent** (03/2024).

China Undergraduate Physics Tournament (CUPT) *12/2021 – 10/2022*
Contestant representing SYSU

- Reviewed literature on fluid mechanics and Hamiltonian mechanics; designed experiments and developed theoretical models for physics problems motivated by everyday phenomena; presented results in the Central South Regional and National rounds.
- Received the **National First Prize** at CUPT (10/2022).
- Received the **First Prize** in the Central South Regional Competition (06/2022).

Extracurricular Activities

SYSU Freshman Mentoring Program *09/2024 – 09/2025*
Mentor

- Mentored a group of incoming students throughout their first year, offering guidance on their studies and the transition to university life.

SYSU Torchwood Association of Physics *10/2022 – 10/2024*
Governing Member

- Organized information-sharing seminars and physics contests, including the CUPT Simulation Game.
- Served as a judge in several contests.

Skills

- **Python:** Self-taught during undergraduate study, with further practice in computational physics coursework. Familiar with core syntax and classes, and experienced in plotting with Matplotlib. Currently learning PyTorch and able to read and understand existing code.
- **Mathematica:** Regular use for physics calculations and review of Mathematica code generated by Claude Code.
- **LaTeX and Markdown:** Proficient in package use and mathematical typesetting; used for notes, assignments, papers, and lab reports.
- **Linux:** Running SMEFTsim workflows in a Linux virtual machine.
- **Bash:** Directory navigation, file operations, and running commands in Linux.
- **SSH:** Configuring SSH keys and remote connections in VS Code.